

Management Strategy Evaluation for Anchovy in ICES Division 9.a South

Working Document

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Introduction

Management context

MSE framework

Operating model

Recruitment conditioning

Recruitment dynamics and stock-recruitment relationship

Recruitment dynamics in the Operating Model (OM) were conditioned using historical spawning stock biomass (SSB) and recruitment estimates derived from the final Stock Synthesis (SS3) assessment for anchovy in Division 9a South (ane.27.9aS).

Recruitment was assumed to occur in the third quarter (Q3), while spawning stock biomass was evaluated in the second quarter (Q2), consistent with the biological timing implemented in the assessment model.

A Beverton-Holt stock-recruitment (SR) relationship was fitted to the historical SSB and recruitment series using the FLSR framework. The fitted OM SR relationship was subsequently compared with the Beverton-Holt parameterization estimated within SS3 to verify consistency between the assessment model and the operating model. Predicted recruitment values from both formulations were compared across the observed range of SSB, and differences in expected recruitment and residuals patterns were quantified.

To facilitate interpretation of recruitment uncertainty, recruitment envelopes corresponding to the estimated recruitment variability were calculated around the SS3 stock-recruitment relationship. Median recruitment predictions were combined with lognormal recruitment deviations using the estimated recruitment standard deviation (σ_R) from the assessment model.

Recruitment residual diagnostics

Historical recruitment residuals were calculated as:

$$\varepsilon_t = \log(R_{obs,t}) - \log(R_{pred,t})$$

where ($R_{obs,t}$) represents observed recruitment and ($R_{pred,t}$) the recruitment predicted by the fitted Beverton-Holt relationship.

The distributional properties of recruitment residuals were evaluated to determine whether the assumptions commonly adopted in fisheries simulated models were supported by the historical data.

Diagnostics included:

- Time-series plots of recruitment residuals
- Histograms and kernel density estimates
- Quantile-quantile (QQ) plots
- Summary statistics including mean, standard deviation, skewness and kurtosis
- Comparison of observed residual quantiles with those expected under a lognormal recruitment process.

Normality of residuals was evaluated using the Shapiro-Wilk test. The objective was not to formally validate a strict normal distribution, but rather to identify substantial departures from the lognormal assumption commonly used in stock assessment and management strategy evaluation frameworks.

Autocorrelation analysis

Regime-shift diagnostics

Development of recruitment scenarios

Results

Historical recruitment dynamics

Historical recruitment exhibited substantial interannual variability throughout the period 1989–2024, ranging approximately from 3.5 to 12 million recruits. No persistent increasing or decreasing trend was evident, and no prolonged periods of consistently low recruitment were observed.

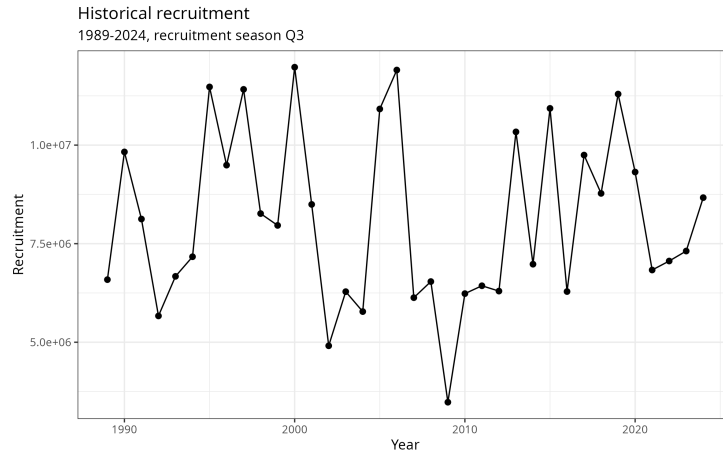


Figure 1: xxx

Stock-recruitment relationship

The Beverton–Holt relationship fitted within the Operating Model closely reproduced the stock–recruitment relationship estimated by the SS3 assessment. Differences between both curves were small across the observed range of spawning biomass, indicating that the OM adequately captures the productivity assumptions of the assessment model.

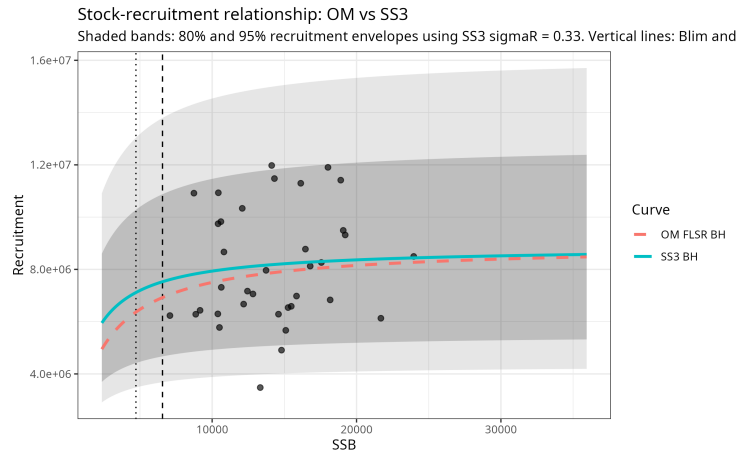


Figure 2: xxx

Table 1: Stock-recruitment parameters used to compare the OM and SS3 Beverton-Holt relationships.

parameter	value	source
a_OM	8926007.024	FLSR OM fit
b_OM	1898.343	FLSR OM fit
R0_SS3	8291985.468	SS3
h_SS3	0.800	SS3
SSB0_SS3	17217.700	SS3
sigmaR_SS3_or_used	0.330	SS3
sigmaR_OM_resid	0.280	OM residuals

Recruitment residuals diagnostics

Recruitment residuals were approximately symmetric and centred around zero. The histogram and QQ plot indicated a reasonable agreement with the assumed lognormal error structure, although some deviations were observed in the tails of the distribution, particularly for a small number of low-recruitment years.

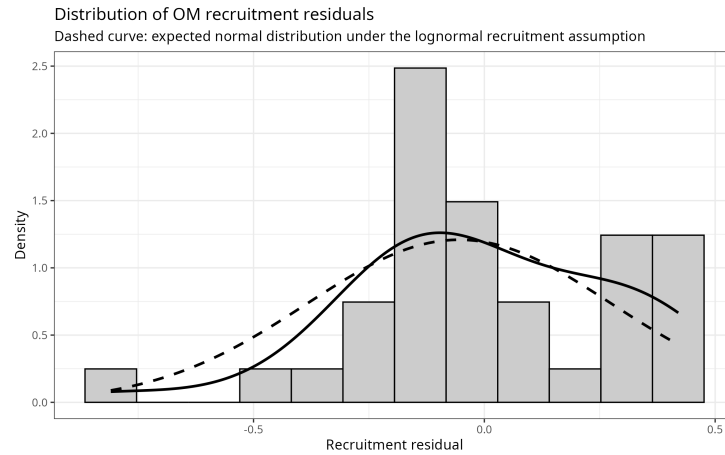


Figure 3: xxx

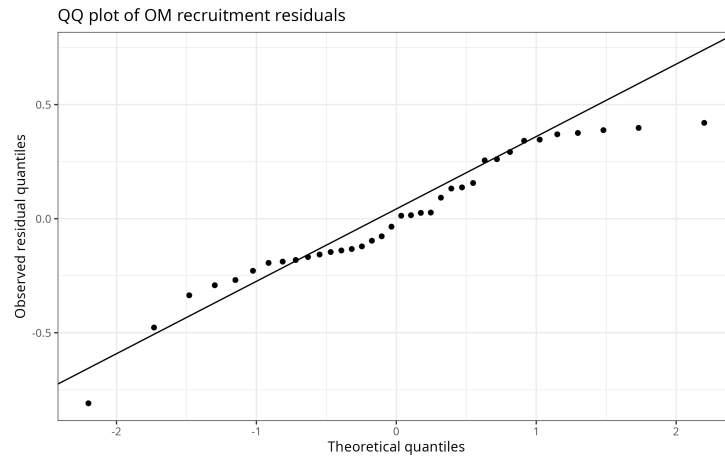


Figure 4: xxx

Table 2: Recruitment residual diagnostic tests.

test	statistic	p_value
Ljung-Box lag 1	0.545	0.460
Ljung-Box lag 3	3.454	0.327
Ljung-Box lag 5	7.656	0.176
Shapiro-Wilk normality	0.946	0.080

Temporal dependence and regimen-shift diagnostics

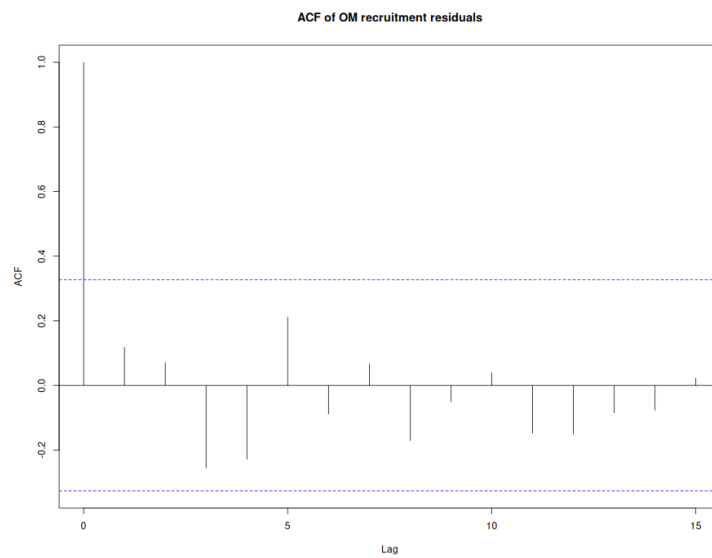


Figure 5: xxx

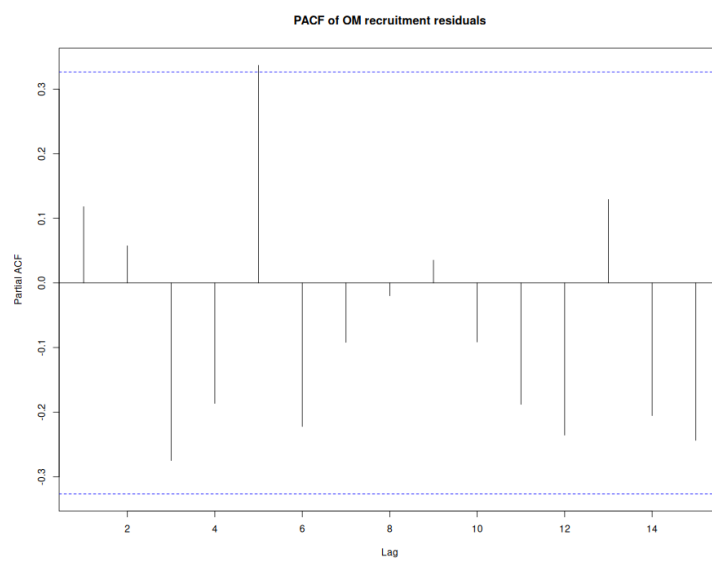


Figure 6: xxx

Table 3: AR(1) model fitted to recruitment residuals.

parameter	estimate	se	z	p_value
ar1	0.117	0.165	0.711	0.477
intercept	0.000	0.052	-0.004	0.997

Table 4: Change-point diagnostics for historical recruitment.

method	n_change_points	change_point_in-dex	change_point_year
mean	0	NA	NA
mean_and_variance	0	NA	NA

Table 5: Structural break diagnostics for historical recruitment.

method	break_index	break_year
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Table 6: Candidate recruitment scenarios for the MSE Operating Model.

OM_scenario	a_multi-SR_model	sig-maR_val	AR1_rho	rationale
base	Beverton-1.00 Holt	0.28	0.00	Reference OM based on fitted BH and observed recruitment residuals.
high_sig-maR_Thorson	Beverton-1.00 Holt	0.74	0.00	High recruitment variability scenario using Thorson et al. 2014 global mean sigmaR.
high_sig-maR_150	Beverton-1.00 Holt	0.42	0.00	Strong increase in recruitment variability relative to ane9aS residuals.
low_productivity_075	Beverton-0.75 Holt	0.28	0.00	Moderate low-productivity regime.
low_productivity_050	Beverton-0.50 Holt	0.28	0.00	Severe low-productivity regime.

OM_sce- nario	a_mul- ti- SR_model	ti- plier	sig- maR_val	AR1_coef	rationale
AR1_re- cruit- ment_Thor- son	Beverton-1.00 Holt	1.00	0.28	0.43	Exploratory scenario using Thorson et al. 2014 mean autocorrelation.
poor_re- cruit- ment_tail	Beverton-1.00 Holt	1.00	0.28	0.00	Stress-test based on observed lower-tail recruitment residuals.

HCR

Scenarios

Performance metrics

Preliminary results